WHITEPAPER · INTEGRATED LENDING OPERATING LAYER (ILOL) V3.0

Governing the Credit Decision: The Case for Governance-Native Risk Infrastructure

"Every decision. Explainable. Auditable. Defensible."

Version 1.0 · April 2026 · Public Distribution · Audience: CRO · VP Decision Science · Model Risk Managers · Compliance Officers

THE PROBLEM

Mid-market lenders ($100M–$10B AUM) face a structural governance crisis. Four dimensions drive $2–15M in SR 11-7 remediation exposure, $500K–$5M in annual exam costs, and fair lending enforcement risk:

Dimension	Pain	Consequence
Decisioning	Fragmented engines, no unified audit trail	SR 11-7 findings, fair lending exposure
Monitoring	30–90 day reporting lag	Slow response to credit deterioration
Compliance	Manual exam prep, 6–12 weeks per package	$500K–$5M exam prep annually
AI Analytics	Generic LLMs hallucinate on regulated data	Unverifiable outputs = regulatory liability

Point solutions don't share data. Legacy platforms produce monthly static reports. In-house builds take 18–24 months. General-purpose AI tools produce answers with no source traceability — a liability in any regulated environment.

THE SOLUTION: ILOL

The **Integrated Lending Operating Layer (ILOL)** is a production-grade, governance-native credit risk platform — a single operating layer across originations, underwriting, compliance, portfolio analytics, and AI-assisted decisioning. All modules share one immutable audit log, one policy registry, and one data lineage graph.

> **Core thesis:** Governance is not overhead — it is a competitive asset. Every ILOL output is traceable to a specific policy version, model artifact, feature snapshot, or source SQL query. By design, not by manual assembly.

SIX INTEGRATED MODULES

Module	What It Delivers
1. Governance Infrastructure	Immutable decision log (SHA-256 hash chain); cryptographically signed policy versioning (RSA-256); continuous AIR/BISG fair lending monitoring; field-level data lineage
2. Decisioning Engine	Sub-200ms REST API; AST-safe policy DSL (no `eval()`); SHAP explainability; PD + fraud scores with Reg B reason codes in one response
3. Portfolio Monitoring	Live approval/delinquency/vintage metrics; PSI + AUC degradation alerts; NLG-powered CRO memos and board packages
4. Compliance & Audit Suite	One-click OCC/CFPB/SR 11-7 exam packet from production artifacts; Reg B adverse action management
5. Model Lifecycle	MLflow-backed registry; auto-generated SR 11-7 model cards; live PSI/AUC/KS monitoring; A/B testing framework
6. RAG AI Analytics Agent	NL query → SQL + Python → BigQuery execution → answer + full source code in UI + stored as immutable audit evidence. Five-layer zero-hallucination enforcement.

ARCHITECTURE SUMMARY

```
LOS → Decision API (Cloud Run / FastAPI)
    → Feature Pipeline
    → XGBoost PD + Fraud Models (SHAP)
    → AST Policy DSL → APPROVE / DECLINE / REFER
    → Immutable Audit DB (Postgres, append-only)
    → BigQuery Warehouse
          ↑
    RAG AI Agent: NL query → schema-bound
    SQL/Python → execute → code surfaced → logged
```

Stack: Python 3.11 · FastAPI · GCP Cloud Run · XGBoost · SHAP · MLflow · PostgreSQL · BigQuery · GCS · Next.js

KEY DIFFERENTIATORS VS. ALTERNATIVES

Capability	ILOL	Best Alternative
Sub-200ms decisioning + SHAP + Reg B	✅	Partial (Zest AI only)
Cryptographic policy versioning + rollback	✅	❌ None
Immutable audit log (hash chain)	✅	❌ None
Continuous fair lending / AIR monitoring	✅	❌ / Manual only
One-click OCC/CFPB exam packet	✅	❌ None
AI analytics with code transparency	✅	❌ No auditability
Auto-generated SR 11-7 model docs	✅	❌ / Manual only
End-to-end field-level data lineage	✅	Partial (dbt only)

MEASURABLE IMPACT (90-DAY TARGETS)

6 hrs	On-demand	Seconds	Live
Exam package generation (was 6–12 wks)	Fair lending analysis (was 3–4 wks + $80K)	Portfolio Q&A (was days of manual joins)	Model docs (was manual, quarterly)

THREE USE CASES IN BRIEF

> **Fair Lending / CFPB Exam:** AI agent runs full ECOA disparate impact analysis (AIR + BISG regression) on 18 months of originations in <30 seconds, with full SQL + Python for independent examiner verification. Previously: $50K–$80K consultant, 3–4 weeks.

> **Policy Simulation:** Analyst simulates a +20pt credit score cutoff change against 180 days of actual decisions in under 1 hour. Promotion requires dual-control authorization; auto-versioned and signed. Previously: 2–3 days manual data pull.

> **OCC Exam Package:** One click generates model inventory, SR 11-7 docs, policy changelog, 200-record decision sample, and fair lending summary — all from production artifacts. Previously: 6–8 weeks across five systems.

AI AGENT: ZERO-HALLUCINATION ARCHITECTURE

1. **Schema Binding** — constrained to live BigQuery schema; no fabricated column names
2. **Result Grounding** — every metric linked to source table + partition + query; ungrounded outputs blocked
3. **Confidence Flagging** — low-confidence answers disclosed, never hallucinated
4. **Session Audit Log** — every query → plan → code → result stored immutably
5. **Human Escalation** — regulatory and board queries require review confirmation

HONEST LIMITATIONS (APRIL 2026)

Current platform gaps — documented accurately:

- **Multi-tenant JWT enforcement** — in schema; not yet enforced at all API handlers (P0 item)
- **Deterministic replay** — components built; end-to-end validation pending
- **Monitoring dashboard** — currently on mock data; BigQuery connection in current sprint
- **No bundled credit models** — ILOL provides runtime; institution supplies PD + fraud models
- **US regulatory scope only** — OCC, CFPB, Fed, FDIC; FCA/OSFI/EBA not yet available
- **AI agent scope** — warehouse data only; no real-time external data lookups
- **Data quality dependency** — output quality scales with upstream data completeness

ROADMAP

Horizon	Items
Q2–Q3 2026	Full multi-tenant enforcement · Bureau integrations (Experian/Equifax/TU) · Open banking enrichment (Plaid/MX) · Production dashboard · Deterministic replay
Q3–Q4 2026	Portfolio P&L engine · CECL reserve modeling · Stress testing · nCino/Blend connectors · AI-proposed alert resolutions (human-approved)
2027+	Continuous learning loop · Multi-institution governance console · Kafka streaming decisioning · International regulatory templates (FCA/OSFI/EBA)

DEPLOYMENT & PILOT MODEL

- **Integration:** Single Decision API REST endpoint on existing LOS — no system replacement
- **Infrastructure:** GCP (Cloud Run, BigQuery, GCS) — customer's own tenant or ILOL-hosted
- **30-day pilot structure:**
  1. Decision API live with customer data by Day 7
  2. Governance artifacts generated from production decisions by Day 21
  3. First exam package exported and reviewed together by Day 30
- **Risk-free clause:** No Year 1 obligation if success criteria are not met by Day 30
- **Year 1 ARR:** $120K–$350K upon pilot conversion

TARGET MARKET

Segment	Profile	ARR Range
Fintech Lenders	$100M–$2B AUM; model-driven decisioning	$150K–$350K
Credit Unions	$200M–$5B assets; SR 11-7 pressure	$150K–$450K
Community Banks	$500M–$10B; OCC/CFPB exam risk	$250K–$600K

TAM: ~4,700 US institutions · Year 1 target: 15–25 customers · Year 2 ARR: $5M–$8M

COMPETITIVE LANDSCAPE

```
                  HIGH GOVERNANCE
                        |
          ┌─────────────┤
          |    [ILOL]   | ← Governance + Automation
          |             |   No competing vendor here
LOW ──────┼─────────────┼──────────────── HIGH
AUTO.     |             |              AUTOMATION
          |             |   [Zest AI] [ML-Only Vendors]
          └─────────────┤
                  LOW GOVERNANCE
```

Alternative	Strength	Gap vs ILOL
Zest AI	ML model quality	No exam automation, no AI agent, no policy versioning
Tableau / Power BI	Visualization	No decisioning, no audit log; AI outputs not submittable as evidence
ChatGPT / Gemini	NL interface	No institution data; hallucination risk; zero audit trail
Jupyter + dbt	Flexibility	Full build-out required; no exam automation; docs go stale
Provenir	LOS integration	No model risk management, no fair lending module, no AI agent
In-house build	Customization	18–24 months to build; talent retention risk; no regulatory templates

WHY NOW

SR 11-7 scrutiny has reached mid-market lenders. CFPB algorithmic adverse action enforcement is intensifying. CECL reserve modeling is a compliance mandate. And general-purpose AI tools are being deployed in regulated contexts without auditability controls.

The technical foundations — XGBoost-grade ML, cloud-native infrastructure, natural language AI — now make it possible to build what was not buildable five years ago: a governance-native platform that operates at sub-200ms latency, maintains comprehensively auditable records, and surfaces AI analytics independently verifiable by a regulator.

The window for first-mover positioning in governance-native credit risk infrastructure for mid-market lenders is open now.

THREE DIAGNOSTIC QUESTIONS

1. If a regulator asked today for every credit decision in the past 18 months — model version, policy version, reason codes — *how long would it take?*
2. If a data scientist asked what caused the approval rate shift in Q3 — *how many systems, how many days?*
3. If the board needs portfolio credit quality for next week's meeting — *how many analyst-hours?*

> ILOL is built to make the answer to all three: **less than a day, from a single system.**

Pilot programs are 30-day engagements with defined success criteria and a risk-free conversion clause. If ILOL does not deliver working governance artifacts from your data in 30 days, there is no Year 1 contract obligation.

ILOL — Integrated Lending Operating Layer · Unified Edition v3.0 · Version 1.0 · April 2026 · Public Distribution

All capability descriptions are grounded in the ILOL Unified PRD v3.0.0 and Technical Architecture Reference v1.0. Maturity levels accurately reflect implementation status as of April 2026. No capability has been represented beyond its documented development state.